Predicting Building Energy Consumption

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1 Problem Statement & Motivation

Buildings account for roughly **40% of total energy consumption** in the United States, and an estimated 20 to 30 percent of that energy is wasted because heating and cooling systems operate on fixed schedules rather than responding to actual demand [Runge and Zmeureanu, 2019]. Predicting hourly energy use at the individual meter level would let facility operators optimize schedules, catch failing equipment early, and decide which buildings to prioritize for retrofits. **Our goal** is to predict meter-level hourly consumption given building metadata and weather, and to determine which class of ML method performs best across different building types and meter types.

This is a challenging prediction problem because of the **heterogeneity** in the data. A university campus, a hotel, and a parking garage produce fundamentally different consumption patterns, and energy use has a **non-linear relationship with temperature**, producing a U-shaped HVAC response that linear models cannot capture without explicit feature engineering.

The ASHRAE Great Energy Predictor III dataset [Miller et al., 2020] provides over 20 million hourly readings from 1,449 buildings across 16 sites, making it the largest open benchmark for building energy prediction. The associated Kaggle competition attracted 3,614 teams; Miller et al. [2022a] found that gradient boosting with careful feature engineering dominated, and that preprocessing mattered more than model architecture. A follow-up error analysis by Miller et al. [2022b] revealed over 60% error on hot water meters, but because it aggregated across many competitors' different pipelines, it cannot isolate model choice from preprocessing differences. Separately, Grinsztajn et al. [2022] show that tree-based models outperform deep learning on tabular data across 45 benchmarks, but do not test this on energy data where domain-specific features like temporal lags and weather interactions play a central role.

We address this gap through a **structured comparison of ten methods** drawn from four complementary model families: linear models, tree ensembles, time series models, and neural networks, all trained on **identical features and the same time-based split**. Each family tests a different modeling assumption about the data. We also test whether **clustering** buildings by consumption profile improves per-group predictions, and perform **per-building-type and per-meter-type failure analysis** to understand not just which method wins overall, but where and why each one breaks down.

2 Dataset & Features

We use the ASHRAE GEPIII dataset from Kaggle.[1] The training set contains **20,216,100 hourly meter readings** from **1,449 buildings** across **16 sites**, covering the year 2016. The data has three files: **train.csv** with `building_id`, `meter` type (electricity, chilled water, steam, hot water), hourly `timestamp`, and `meter_reading` in kWh as the regression target; **building_metadata.csv** with site, primary use across 16 categories, square footage, year built (60% missing), and floor count (75% missing); and **weather_train.csv** with hourly temperature, dew point, wind speed, cloud coverage (40% missing), and precipitation per site. The target is heavily right-skewed, so we train on $\log(1+y)$

[1] `https://www.kaggle.com/competitions/ashrae-energy-prediction`

and evaluate on the original scale. For experiments we use a 3-to-5 site subset of roughly 1 to 3 million rows.

We engineer approximately **28 features** designed to capture the main drivers of building energy use: temporal encodings with cyclical hour and day components, holiday and weekend flags, since building occupancy follows strong daily and weekly cycles; lag and rolling statistics per meter to capture recent consumption trends; weather features including temperature squared for the U-shaped HVAC response, where both extreme cold and heat increase demand; building metadata; interaction terms; and target encoding computed only within the training fold. All features use **strictly past data**. We split **January through September** for training and **October through December** for validation with no shuffling.

3 Methods

Our approach is to start with the simplest possible model and progressively relax assumptions, so that each step reveals what the previous one was missing. All ten models are trained on the same 28 features and time-based split to ensure a fair comparison.

We begin with **linear models** to test whether the relationship between features and consumption is approximately additive: OLS as an unregularized baseline, Ridge for L2 regularization, Lasso for L1 feature selection, and ElasticNet combining both. We sweep λ via time-series cross-validation.

Where linear models fall short, **tree-based models** can capture non-linear interactions automatically. A single decision tree serves as an interpretable baseline, Random Forest reduces variance through bagging, and LightGBM [Ke et al., 2017] further reduces bias through sequential boosting. LightGBM was the dominant method in the ASHRAE competition [Miller et al., 2022a].

Since energy data is inherently temporal, we also evaluate **time series models** to test whether explicitly modeling sequence dependence helps beyond hand-crafted lag features. Per-meter ARIMA serves as a classical baseline, well suited here because building energy exhibits strong daily and weekly periodicity; we fit ARIMA only on the site subset and use `auto_arima` to keep fitting tractable across hundreds of meters. An LSTM with a 168-hour lookback, corresponding to one full week, tests whether recurrent architectures can learn patterns that lag features miss.

Finally, a **3-layer MLP** with 256, 128, and 64 hidden units, ReLU, dropout, and early stopping tests deep learning on tabular features directly. Based on Grinsztajn et al. [2022], we hypothesize it will not outperform LightGBM, and diagnosing why is part of our analysis.

As a complementary unsupervised perspective, we also run a **clustering experiment**: we compute each building's normalized average consumption by hour of day, then group buildings using K-means with K selected via the elbow method. We train per-cluster LightGBM models to test whether reducing heterogeneity improves accuracy.

4 Evaluation & Analysis

The goal of our evaluation is not just to rank models by accuracy, but to understand when and why each family succeeds or fails. We measure **RMSE** on the original scale as the primary metric, **CV-RMSE** for normalized cross-building comparison [Runge and Zmeureanu, 2019], **MAE** as a secondary metric, and wall-clock training time.

Our analysis has four components. First, we run all models on 5 raw features and again on the full 28 engineered features to isolate the effect of **feature engineering versus model choice**. We expect the gap between linear models and LightGBM to shrink substantially with better features, replicating the competition's finding that preprocessing matters more than architecture [Miller et al., 2022a]. Second, we compare ARIMA and LSTM against non-temporal models that use hand-crafted lags to evaluate whether **explicit temporal modeling** adds value. We hypothesize that well-engineered lag features capture most temporal structure, making the sequence models only marginally better or possibly worse due to higher variance. Third, we break down RMSE by building use and meter type to identify **per-building-type failure modes**. We expect Education buildings with regular weekday schedules to be well-predicted by all methods, while Lodging and Healthcare with irregular around-the-clock operation should expose where simpler models fail. Fourth, we compare global models against per-cluster models to measure the **clustering benefit**, expecting the largest gains at sites with diverse building mixes where a global model must compromise across very different consumption patterns.

Finally, we characterize the remaining performance gap by examining the worst-predicted buildings and meters to identify what information is missing from the dataset, such as occupancy schedules or equipment state, and what additional data sources would be needed to close it.

5 Challenges and Mitigations

Temporal leakage. We enforce a strict chronological split and unit-test that no feature at time t uses future data.

Missing data. `year_built` at 60% missing is imputed with site medians; `floor_count` at 75% missing is dropped. Weather gaps are filled by linear interpolation.

Building-type imbalance. Education is 41% of the data; Healthcare and Parking under 2% each. We report per-building-type RMSE as a standard part of all results and evaluate whether training separate per-type models improves accuracy on underrepresented categories.

References

L. Grinsztajn, E. Oyallon, and G. Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems*, 2022.

G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.

C. Miller et al. The ASHRAE Great Energy Predictor III competition: Overview and results. *Sci. Technol. Built Environ.*, 26(10):1427–1447, 2020.

C. Miller, L. Hao, and C. Fu. Gradient boosting machines and careful pre-processing work best: ASHRAE GEPIII lessons. *ASHRAE Trans.*, 128:405–413, 2022.

C. Miller, B. Picchetti, C. Fu, and J. Pantelic. Limitations of machine learning for building energy prediction: ASHRAE GEPIII error analysis. *Sci. Technol. Built Environ.*, 28(8):1036–1052, 2022.

J. Runge and R. Zmeureanu. Forecasting energy use in buildings using artificial neural networks: A review. *Energies*, 12(18):3355, 2019.